



Aviation Investigation Final Report

Location:	Colusa, California	Accident Number:	WPR24LA299
Date & Time:	August 30, 2024, 07:05 Local	Registration:	N710WD
Aircraft:	WILLIAMS HELICOPTER CORP UH-1H	Aircraft Damage:	Substantial
Defining Event:	Loss of control in flight	Injuries:	1 Serious
Flight Conducted Under:	Part 137: Agricultural		

Analysis

The pilot of the helicopter reported that while repositioning closer to the loading equipment, the helicopter experienced collective bounce. He attempted to maneuver the helicopter to avoid equipment and personnel; however, control was lost and the helicopter impacted terrain.

The operator reported that there were no preaccident mechanical malfunctions or failures that would have precluded normal operations. The operators manual for the helicopter stated, in part, *“Collective bounce is a pilot Induced vertical oscillation of the collective control system when an absolute friction (either pilot applied, or control rigged) is less than seven pounds. It may be encountered in any flight condition by a rapid buildup of vertical bounce at approximately three cycles per second. The severity of the oscillation is such that effective control of the helicopter may become difficult to maintain. The pilot should apply and maintain adequate collective friction in all flight conditions.”*

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be:

The pilot’s failure to set collective friction which resulted in collective bounce and subsequent loss of control.

Findings

Personnel issues	Use of equip/system - Pilot
Personnel issues	Aircraft control - Pilot

Factual Information

History of Flight

Maneuvering-hover	Loss of control in flight (Defining event)
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Pilot Information

Certificate:	Commercial	Age:	60,Male
Airplane Rating(s):	Single-engine land; Single-engine sea; Multi-engine land	Seat Occupied:	Right
Other Aircraft Rating(s):	Helicopter	Restraint Used:	4-point
Instrument Rating(s):	Airplane	Second Pilot Present:	No
Instructor Rating(s):	None	Toxicology Performed:	
Medical Certification:	Class 2 Without waivers/limitations	Last FAA Medical Exam:	February 6, 2024
Occupational Pilot:	Yes	Last Flight Review or Equivalent:	October 11, 2023
Flight Time:	13000 hours (Total, all aircraft), 22 hours (Total, this make and model), 13000 hours (Pilot In Command, all aircraft), 125 hours (Last 90 days, all aircraft), 50 hours (Last 30 days, all aircraft), 1 hours (Last 24 hours, all aircraft)		

Aircraft and Owner/Operator Information

Aircraft Make:	WILLIAMS HELICOPTER CORP	Registration:	N710WD
Model/Series:	UH-1H	Aircraft Category:	Helicopter
Year of Manufacture:	1965	Amateur Built:	
Airworthiness Certificate:	Restricted (Special)	Serial Number:	65-09761
Landing Gear Type:	Skid	Seats:	2
Date/Type of Last Inspection:	AAIP	Certified Max Gross Wt.:	9500 lbs
Time Since Last Inspection:		Engines:	1 Turbo shaft
Airframe Total Time:		Engine Manufacturer:	Ozark-Aeroworks
ELT:	Installed, not activated	Engine Model/Series:	T53-L-13B
Registered Owner:	A&P HELICOPTERS INC	Rated Power:	1318 Horsepower
Operator:	A&P HELICOPTERS INC	Operating Certificate(s) Held:	Rotorcraft external load (133), Agricultural aircraft (137)

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual (VMC)	Condition of Light:	Day
Observation Facility, Elevation:	KOVE, 187 ft msl	Distance from Accident Site:	23 Nautical Miles
Observation Time:	07:53 Local	Direction from Accident Site:	49°
Lowest Cloud Condition:	Clear	Visibility	10 miles
Lowest Ceiling:	None	Visibility (RVR):	
Wind Speed/Gusts:	3 knots / None	Turbulence Type Forecast/Actual:	None / None
Wind Direction:	130°	Turbulence Severity Forecast/Actual:	N/A / N/A
Altimeter Setting:	30.01 inches Hg	Temperature/Dew Point:	21°C / 12°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Richvale, CA (07CL)	Type of Flight Plan Filed:	None
Destination:	Richvale, CA (07CL)	Type of Clearance:	None
Departure Time:	06:40 Local	Type of Airspace:	Class G

Wreckage and Impact Information

Crew Injuries:	1 Serious	Aircraft Damage:	Substantial
Passenger Injuries:	N/A	Aircraft Fire:	None
Ground Injuries:		Aircraft Explosion:	None
Total Injuries:	1 Serious	Latitude, Longitude:	39.247946,-121.99082

Administrative Information

Investigator In Charge (IIC):	Salazar, Fabian
Additional Participating Persons:	Brook Stewart; Federal Aviation Administration; Sacramento, CA
Original Publish Date:	April 22, 2025
Last Revision Date:	
Investigation Class:	Class 4
Note:	The NTSB did not travel to the scene of this accident.
Investigation Docket:	https://data.nts.gov/Docket?ProjectID=195085

The National Transportation Safety Board (NTSB) is an independent federal agency charged by Congress with investigating every civil aviation accident in the United States and significant events in other modes of transportation—railroad, transit, highway, marine, pipeline, and commercial space. We determine the probable causes of the accidents and events we investigate, and issue safety recommendations aimed at preventing future occurrences. In addition, we conduct transportation safety research studies and offer information and other assistance to family members and survivors for each accident or event we investigate. We also serve as the appellate authority for enforcement actions involving aviation and mariner certificates issued by the Federal Aviation Administration (FAA) and US Coast Guard, and we adjudicate appeals of civil penalty actions taken by the FAA.

The NTSB does not assign fault or blame for an accident or incident; rather, as specified by NTSB regulation, “accident/incident investigations are fact-finding proceedings with no formal issues and no adverse parties ... and are not conducted for the purpose of determining the rights or liabilities of any person” (Title 49 *Code of Federal Regulations* section 831.4). Assignment of fault or legal liability is not relevant to the NTSB’s statutory mission to improve transportation safety by investigating accidents and incidents and issuing safety recommendations. In addition, statutory language prohibits the admission into evidence or use of any part of an NTSB report related to an accident in a civil action for damages resulting from a matter mentioned in the report (Title 49 *United States Code* section 1154(b)). A factual report that may be admissible under 49 *United States Code* section 1154(b) is available [here](#).